

Sensorless Position Control for Surface Permanent Magnet Synchronous Motors at Zero Speed and Acceleration

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Abstract: This paper proposes a position, speed and load torque observation technique for position control of a non-salient Permanent Magnet Synchronous Motor (PMSM). Unlike the speed control, the position control implies that the motor is often in low speed and standstill conditions. Observability analyze is done to show that the motor is not observable in standstill condition. A high-frequency signal injection is introduced to get round this unobservability. An Extended Kalman Filter is then used to observe the rotor position, speed and load torque by taking into account the injected signal into the observation model. Position tracking and load torque influence are studied on a test bench. Experimental results are provided to confirm the performance and efficiency of the proposed solution.

1. INTRODUCTION

Permanent Magnet Synchronous Motors are widely used in industry due to their very high efficiency and power density. Especially in aeronautics, PMSM are used in electro-mechanical actuator positioning, including a flap actuator. For this kind of application, surface PMSM is preferred than a salient one due to compactness constraint. To allow highly efficient control of PMSM, a good knowledge of the electrical position of the rotor is crucial. Therefore, motor position and speed monitoring or mechanical sensorless control can eventually increase the reliability of the system. Unfortunately, the techniques based on the model of the motor by using estimator (Wu *et al.*, 1991) or observers (Kim *et al.*, 1997) (Bolognani *et al.*, 2003) (Ilioudis *et al.*, 2008), (Foo *et al.*, 2010) can't give satisfactory results for position control because the system is not observable at standstill condition (Zaltni *et al.*, 2009). Others techniques which explore the saliency of the machine by injecting high frequency signal are not effective for surface (not salient) PMSM (Foo *et al.*, 2010), (Consoli *et al.*, 2008). A review of these methods can be found in (Benjak *et al.*, 2010, 1,2,3).

In (Abry *et al.*, 2011) we have proposed an approach combining a signal injection and a model-based method to observe the position and speed of a salient PMSM for all range of the speed including low speed and standstill operation conditions. As the load torque is unknown and influences the control performance, the objective of this paper is to extend the proposed method by adding the load torque estimation. In the first part of this paper, we propose a theoretical observability study of a surface PMSM model extended to load torque. The unobservability at standstill condition is demonstrated. As in (Abry *et al.*, 2011), a dedicated injection signal is introduced. This signal creates the necessary motion that makes the model observable including standstill condition. We thus proposed in section 3

an Extended Kalman Filter for the estimation of the rotor position, speed and load torque using the model with taking into account the injected signal. Experimental validation tests are performed on a test bench. The obtained results are showed in section 4.

2. SURFACE PMSM OBSERVABILITY ANALYSIS

2.1 Modelling

Surface PMSM consists of three phase stator windings and permanent magnets mounted on the rotor surface. This machine can be classically modelled by the electrical equations written in the static Concordia reference frame (α - β) as:

$$\begin{aligned}\frac{dI_\alpha}{dt} &= -\frac{R_s}{L_s}I_\alpha + \frac{\varphi_f}{L_s}\omega_e \sin(\theta) + \frac{1}{L_s}V_\alpha \\ \frac{dI_\beta}{dt} &= -\frac{R_s}{L_s}I_\beta - \frac{\varphi_f}{L_s}\omega_e \cos(\theta) + \frac{1}{L_s}V_\beta\end{aligned}\quad (1)$$

V_α , V_β , I_α and I_β are α - β axis stator voltages and currents respectively, R_s and L_s are the stator phase resistance and inductance respectively, ω_e and θ are the rotor electrical speed and position, φ_f is the flux established by rotor permanent magnets.

The motion equation of the PMSM is shown in (2).

$$\begin{aligned}\frac{d\theta}{dt} &= \omega_e \\ \frac{d\omega_e}{dt} &= \frac{3}{2} \frac{P^2}{J} \varphi_f (\cos(\theta)I_\beta - \sin(\theta)I_\alpha) - \frac{f_v}{J}\omega_e - \frac{PT}{J}\end{aligned}\quad (2)$$

J is the rotor inertial, P is the number of pole pairs, f_v is the viscous friction coefficient and T is the load torque.

The variation of the load torque is usually needed to improve the control performance of PMSM. We propose an extended model (3) which allows estimating the speed, position and load torque from the measured voltages and currents.

$$\begin{cases} \dot{x} = f(x) + Bu \\ y = h(x) = [h_1 \ h_2]^T = Cx \end{cases} \quad (3)$$

with $x = [I_\alpha \ I_\beta \ \theta \ \omega_e \ T]^T$; $y = [I_\alpha \ I_\beta]^T$; $u = [V_\alpha \ V_\beta]^T$

$$f(x) = \begin{bmatrix} -\frac{R_s}{L_s} I_\alpha + \omega_e \frac{\varphi_f}{L_s} \sin(\theta) \\ -\frac{R_s}{L_s} I_\beta - \omega_e \frac{\varphi_f}{L_s} \cos(\theta) \\ \omega_e \\ \frac{1}{J} \left(\frac{3}{2} P^2 \varphi_f (\cos(\theta) I_\beta - \sin(\theta) I_\alpha) - f_v \omega_e - PT \right) \\ 0 \end{bmatrix};$$

$$B = \begin{bmatrix} \frac{1}{L_s} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{L_s} & 0 & 0 & 0 \end{bmatrix}^T; \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

The observability analysis of the non-linear system (3) can be performed by using the rank criterion developed in (Hermann et al., 1977). A sufficient weak local observability condition of (3) is that the rank of the observability matrix O defined as:

$$O = [dh_1 \ dh_2 \ dL_f h_1 \ dL_f h_2 \ dL_f^2 h_1 \ dL_f^2 h_2]^T \quad (4)$$

is equal to the state vector dimension; $dL_f^i h$ is the i^{th} order of Lie derivative of the function h with respect to the vector field f .

The first 5 rows of observability matrix O give :

$$O_{1-5} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -\frac{R_s}{L_s} & 0 & \frac{\varphi_f \omega_e}{L_s} \cos(\theta) & \frac{\varphi_f}{L_s} \sin(\theta) & 0 \\ 0 & -\frac{R_s}{L_s} & \frac{\varphi_f \omega_e}{L_s} \sin(\theta) & -\frac{\varphi_f}{L_s} \cos(\theta) & 0 \\ \dots & \dots & \dots & \dots & -\frac{P\varphi_f}{L_s J} \sin(\theta) \end{bmatrix}$$

and the 1^{st} to 4^{th} and the 6^{th} rows give:

$$O_{1-4,6} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -\frac{R_s}{L_s} & 0 & \frac{\varphi_f \omega_e}{L_s} \cos(\theta) & \frac{\varphi_f}{L_s} \sin(\theta) & 0 \\ 0 & -\frac{R_s}{L_s} & \frac{\varphi_f \omega_e}{L_s} \sin(\theta) & -\frac{\varphi_f}{L_s} \cos(\theta) & 0 \\ \dots & \dots & \dots & \dots & \frac{P\varphi_f}{L_s J} \cos(\theta) \end{bmatrix}$$

The corresponding determinants are:

$$\Delta_{1-5} = +\frac{P\varphi_f^3 \omega_e}{L_s^3 J} \sin(\theta) \quad \text{and} \quad \Delta_{1-4,6} = -\frac{P\varphi_f^3 \omega_e}{L_s^3 J} \cos(\theta) \quad (5)$$

One of these determinants is not zero for $\omega_e \neq 0$. Then a sufficient observability condition can be fulfilled if the motor operates away from zero speed. Higher order Lie derivatives of the output overcoming the observability singularity have been investigated. It shows that the model (3) is not observable when the velocity and acceleration are both zero. Therefore, no information can be derived from the model in this case.

2.2 High frequency signal injection

To get round this unobservability at zero speed and acceleration, we propose to superimpose an extra high frequency (HF) signal to the control signal. The basic idea is to use the injected signal to provide information about the position without having any mechanical effect once the rotor's location is assessed. For these reasons, the following signal is chosen to be injected:

$$\begin{bmatrix} \hat{V}_d \\ \hat{V}_q \end{bmatrix} = \begin{bmatrix} A_i \cos(\omega_i t) \\ 0 \end{bmatrix} \quad (6)$$

where A_i and ω_i are the magnitude and the frequency respectively of the injected signal. $\hat{V}_d$ and $\hat{V}_q$ refer to the stator voltages in an estimated d - q reference frame. It is the result of Park transform at the position $\hat{\theta} = \theta + d\theta$ where θ stands for the real position and $d\theta$ for the position estimation error (Fig.1).

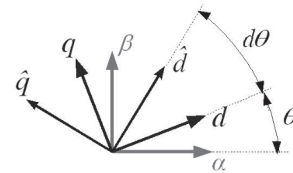


Fig. 1 : Estimated reference frame

A key step is the processing through the Park transform, which uses the estimated position. This transformation is achieved by the following operation:

$$\begin{bmatrix} \hat{V}_d \\ \hat{V}_q \end{bmatrix} = \begin{bmatrix} \cos(d\theta) & \sin(d\theta) \\ -\sin(d\theta) & \cos(d\theta) \end{bmatrix} \begin{bmatrix} V_d \\ V_q \end{bmatrix} \quad (7)$$

In the real Park reference frame, the injected signal is:

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = \begin{bmatrix} A_i \cos(\omega_i t) \cos(d\theta) \\ A_i \cos(\omega_i t) \sin(d\theta) \end{bmatrix} \quad (8)$$

Using a “high-frequency” model of the machine as presented in (Corley *et al.*, 1998), the voltage in d - q reference frame can be simplified to:

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = \begin{bmatrix} L_s s & 0 \\ 0 & L_s s \end{bmatrix} \begin{bmatrix} I_d \\ I_q \end{bmatrix} \quad (9)$$

where s is Laplace operator.

Substituting (8) into (9), the currents generated by the injection can be calculated:

$$\begin{bmatrix} I_d \\ I_q \end{bmatrix} = \frac{A_i \sin(\omega_i t)}{L_s \omega_i} \begin{bmatrix} \cos(d\theta) \\ \sin(d\theta) \end{bmatrix} \quad (10)$$

And the generated torque, proportional to the I_q current, has the following form:

$$T = \frac{3}{2} P \frac{A_i \varphi_f}{L_s \omega_i} \sin(d\theta) \sin(\omega_i t) \quad (11)$$

When the position is correctly estimated ($d\theta=0$), the injection has no effect on the torque. However, when the position is wrongly estimated, the voltage injected creates a torque that itself creates a vibration at the injection frequency. When the position error is increasing, I_q current and torque is increasing sharply, thus creating a movement that provides enough information for the position to be estimated.

2.3 New model with signal injection

The main originality of the proposed method is to consider the injected signal as an intrinsic part of the motor behaviour and not as an external control signal. Therefore by transformed $\hat{V}_d$ and $\hat{V}_q$ in α - β reference frame (Fig.1), a new observation model of surface PMSM taken into account the injected signal can be inferred as:

$$\begin{cases} \dot{x} = f_i(x) + Bu \\ y = Cx \end{cases} \quad (12)$$

with

$$f_i(x) = \begin{bmatrix} -\frac{R_s}{L_s} I_\alpha + \omega_e \frac{\varphi_f}{L_s} \sin(\theta) + \frac{A_i}{L_s} \cos(\omega_i t) \cos(\theta + d\theta) \\ -\frac{R_s}{L_s} I_\beta - \omega_e \frac{\varphi_f}{L_s} \cos(\theta) + \frac{A_i}{L_s} \cos(\omega_i t) \sin(\theta + d\theta) \\ \omega_e \\ \frac{1}{J} \left(\frac{3}{2} P^2 \varphi_f (\cos(\theta) I_\beta - \sin(\theta) I_\alpha) - f_v \omega_e - PT \right) \\ 0 \end{bmatrix}$$

The associated observability matrix becomes:

$$O' = \begin{bmatrix} dh_1 & dh_2 & dL_{f1} h_1 & dL_{f1} h_2 & dL_{f1}^2 h_1 & dL_{f1}^2 h_2 \end{bmatrix}^T$$

Consider the subspace generated from the 5 first rows of O' :

$$O'_{1-5} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -\frac{R_s}{L_s} & 0 & \frac{\varphi_f \omega_e}{L_s} \cos(\theta) - \frac{A_i}{L_s} \cos(\omega_i t) \sin(\theta + d\theta) & \frac{\varphi_f}{L_s} \sin(\theta) & 0 \\ 0 & -\frac{R_s}{L_s} & \frac{\varphi_f \omega_e}{L_s} \sin(\theta) + \frac{A_i}{L_s} \cos(\omega_i t) \cos(\theta + d\theta) & -\frac{\varphi_f}{L_s} \cos(\theta) & 0 \\ \dots & \dots & \dots & \dots & -\frac{P\varphi_f}{L_s J} \sin(\theta) \end{bmatrix}$$

and the subspace generated from the 4 first and the 6th rows of O' :

$$O'_{1-4,6} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -\frac{R_s}{L_s} & 0 & \frac{\varphi_f \omega_e}{L_s} \cos(\theta) - \frac{A_i}{L_s} \cos(\omega_i t) \sin(\theta + d\theta) & \frac{\varphi_f}{L_s} \sin(\theta) & 0 \\ 0 & -\frac{R_s}{L_s} & \frac{\varphi_f \omega_e}{L_s} \sin(\theta) + \frac{A_i}{L_s} \cos(\omega_i t) \cos(\theta + d\theta) & -\frac{\varphi_f}{L_s} \cos(\theta) & 0 \\ \dots & \dots & \dots & \dots & \frac{P\varphi_f}{L_s J} \cos(\theta) \end{bmatrix}$$

Computing the corresponding determinant gives:

$$\Delta'_{1-5} = \left(-\frac{\varphi_f^2}{L_s^2} \omega_e + \frac{\varphi_f}{L_s^2} A_i \cos(\omega_i t) \sin(d\theta) \right) \left(-\frac{P\varphi_f}{L_s J} \sin(\theta) \right)$$

$$\Delta'_{1-4,6} = \left(-\frac{\varphi_f^2}{L_s^2} \omega_e + \frac{\varphi_f}{L_s^2} A_i \cos(\omega_i t) \sin(d\theta) \right) \left(\frac{P\varphi_f}{L_s J} \cos(\theta) \right)$$

These determinants are not zero at the same time when the estimation error $d\theta$ and/or the speed ω_e is not zero. Higher order derivatives of the output overcoming the observability singularity show that the model with injection (12) is locally weakly observable when the velocity, acceleration and position estimation error are not zero at the same time. When the velocity, acceleration are zero at the same time, this observability is guaranteed by the injections which makes weak movement of the rotor and provides position estimation error.

3. PROPOSED ESTIMATION METHOD

The results of observability analysis of previous section show that the model that takes into account the injection of (6) is observable overall operating speed range even at standstill. Consequently we can use an observer to estimate the position, speed and load torque for position control where speed and acceleration are often stated at low values.

As model (12) is nonlinear, a discrete-time Extended Kalman Filter (EKF) was chosen to perform the estimation.

After a discretization and adding process and measurement disturbance w and v , the model (12) becomes:

$$\begin{cases} x_k = f_{i,k-1}(x_{k-1}) + Bu_{k-1} + w_{k-1} \\ y_k = Cx_k + v_k \end{cases} \quad (13)$$

The noise covariance matrices are defined as follows:

$$Q_k = \text{cov}(w_k w_k^t) \quad \text{and} \quad R_k = \text{cov}(v_k v_k^t)$$

The EKF algorithm is computed by iteration as follows (Simon, 2006):

- Compute the partial derivation matrix $F_{k-1} = \left. \frac{\partial f_{i,k-1}}{\partial x} \right|_{\hat{x}_{k-1}^+}$

- Perform the time update of the estimation-error covariance and state estimate $\hat{x}_k^-$ as follows:

$$\begin{cases} P_k^- = F_{k-1} P_{k-1}^- F_{k-1}^t + Q_{k-1} \\ \hat{x}_k^- = f_{i,k-1}(\hat{x}_{k-1}^-) + Bu_{k-1} \end{cases}$$

- Perform the measurement update of the state estimation $\hat{x}_k^+$ and estimation-error covariance as follows:

$$\begin{aligned} K_k &= P_k^- C^t (C P_k^- C^t + R_k)^{-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - C \hat{x}_k^-) \\ P_k^+ &= (I - K_k C) P_k^- (I - K_k C)^t + K_k R_k K_k^t \end{aligned}$$

The covariance matrices Q_k , R_k and the initial value of estimation-error covariance P_0^+ were derived using principle presented in (Bolognani et al., 2003) and adapted to the particular case of the load torque observation and injection. The values used for the filter are:

$$Q_k = \begin{bmatrix} 0.023I_n & 0 & 0 & 0 & 0 \\ 0 & 0.023I_n & 0 & 0 & 0 \\ 0 & 0 & 0.01 & 0 & 0 \\ 0 & 0 & 0 & 0.0046\Omega_n & 0 \\ 0 & 0 & 0 & 0 & 100T_n \end{bmatrix}$$

$$R_k = \begin{bmatrix} 0.023I_n & 0 \\ 0 & 0.023I_n \end{bmatrix}$$

$$\text{and } P_0^+ = \begin{bmatrix} 0.04I_n & 0 & 0 & 0 & 0 \\ 0 & 0.04I_n & 0 & 0 & 0 \\ 0 & 0 & 10 & 0 & 0 \\ 0 & 0 & 0 & 71 \times 10^{-6} \Omega_n & 0 \\ 0 & 0 & 0 & 0 & 0.1T_n \end{bmatrix}$$

where I_n , T_n and Ω_n are rated values of stator current, torque and mechanical speed respectively.

The Fig. 2 summarizes the proposed observer scheme.

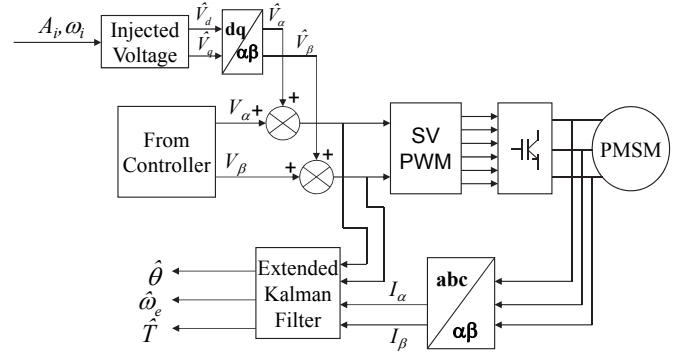


Fig. 1 : Scheme of the proposed estimation method

4. EXPERIMENTAL RESULTS

This proposed position control has been tested on a 1.6kW PMSM with a 4096-pulse incremental encoder solely used to validate the observer results (Fig. 3). Another identical PMSM is used as load torque generator. The characteristics of the PMSM are given in Tab. I. A 15kW commercial inverter is supplied with a DC voltage generator. The current are measured using three LEM current sensors (LEM LA 100P). The proposed control algorithm is performed by a dSpace dS1104 controller board, using Simulink and the Real-time workshop toolbox of MATLAB. The sampling period is fixed to 0.2ms. The injection signal has a 30V magnitude and a 400Hz frequency.

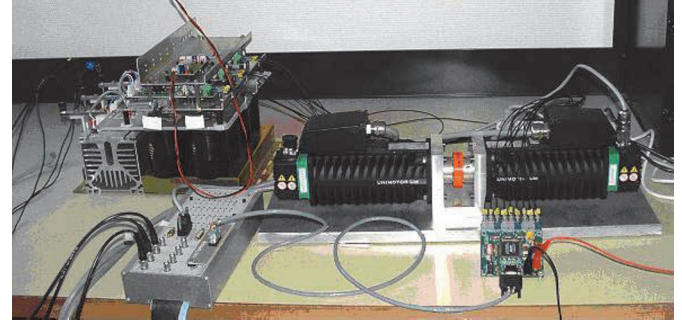


Fig. 2 : Test Bench

Table 1. Motor Parameters

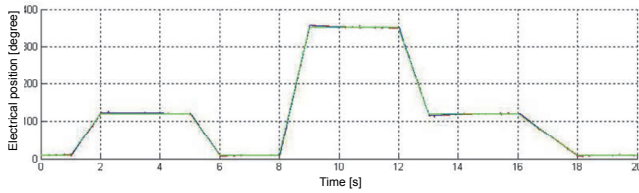
Rated power (Pn)	1.6 kW
Number of pole pairs (P)	3
Stator resistance (R _s)	2.06 Ω
Stator inductance (L _s)	9.15 mH
Inertia (J)	0.00747 kg.m ²
Viscous friction coefficient (f)	0.0249 Nm/rad/s
Permanent magnet flux (φ _p)	0.29 Wb
Rated torque (Tn)	5.09 Nm
Rated current (In)	5.86 A
Rated mechanical speed (Ωn)	3000 rpm

4.1 Position control

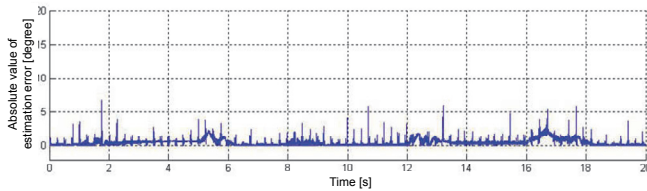
A classical PI controller is designed to track a reference position varying between 0 to 360°. No load is applied ($T=0$).

After an initial position estimation using a magnetic saturation method as (Hu *et al.*, 2009), the observer performances are tested in “monitoring” operation and in “sensorless” operation. The first uses the position sensor for the position control and Park transformation. The observed states are compared with that of sensor. This test is interesting for a position control monitoring application. In “sensorless” operation, the observed rotor speed and position are used in position control and Park transformation. This corresponds to sensorless position control applications. The values measured from position sensor are only used for comparative studies.

Fig. 4 (a) shows the electrical position tracking for successive motor position references in “monitoring” operation. Fig.4 (b) gives the corresponding error between the measured and the estimated electrical positions. It can be seen that the estimation error stays below 5 electrical degrees (less than 2 mechanical degrees) even during very fast transitional states and shows almost no bias at standstill.



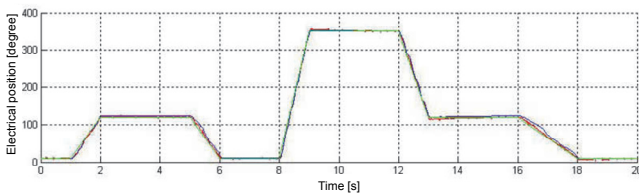
(a) Electrical position tracking: reference position (green), measured position (blue) and estimated position (red)



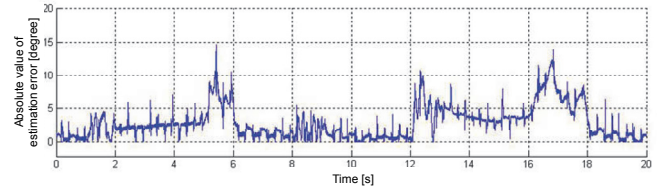
(b) Electrical position error.

Fig. 4 : “Monitoring” test for successive position references

Fig.5 (a) and (b) show respectively the electrical position response and the position estimation error in “sensorless” operation. The sensorless algorithm is able to follow the desired reference and the error remains below 10 electrical degrees (less than 4 mechanical degrees), even during transient state. During steady state, the error is even inferior to 5 electrical degrees.



(a) Electrical position tracking: reference position (green), measured position (blue) and estimated position (red)



(b) Electrical position error.

Fig. 5 : “Sensorless” test for successive position references

4.2 Load torque influence

In order to test the observer robustness versus load torque disturbance, a load torque is applied suddenly to the rotor while this last is controlled at a defined position (Fig. 6).

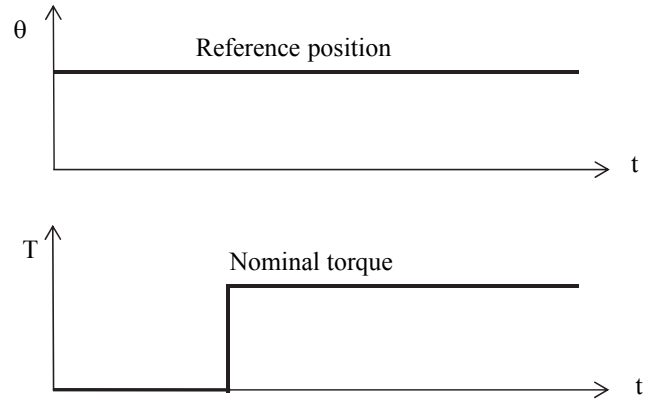
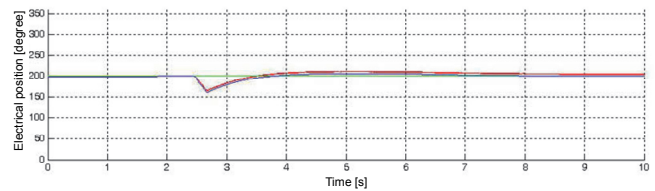
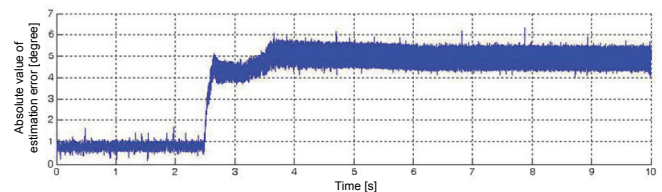


Fig. 6: Load torque application

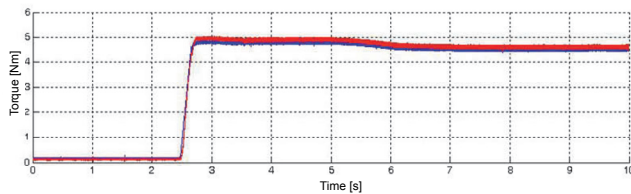
Fig. 7 (a) shows the electrical position tracking in open-loop operation condition. Fig. 7 (b) gives the error between the measured and the estimated electrical positions. The estimation error increases when the load torque applied. However it stays below 6 electrical degrees (less than 2 mechanical degrees) even at standstill. In Fig. 7 (c), load torque calculated from the measured currents and that estimated are compared. It can be seen that the torque is estimated well.



(a) Electrical position tracking: reference position (green), measured position (blue) and estimated position (red)



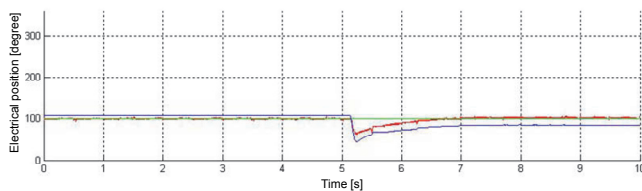
(b) Electrical position error.



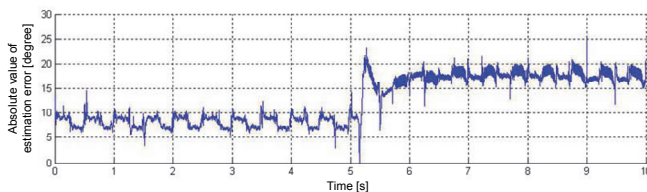
(c) Load torque evolution: calculated from measured currents (blue), estimated (red)

Fig. 7 “Monitoring” test for load torque change

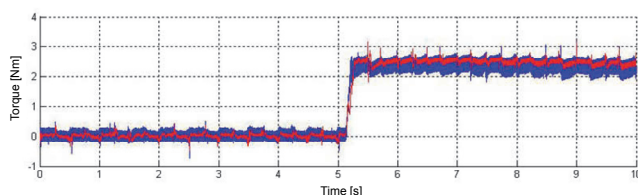
In “sensorless” operation condition, a static error less than 20 electrical degrees (less than 7 mechanical degrees) be found (Fig. 8 b) when the load torque is applied. The load torque is well estimated (Fig. 8.c).



(a) Electrical position tracking: reference position (green), measured position (blue) and estimated position (red)



(b) Electrical position error.



(c) Load torque evolution: calculated from measured currents (blue), estimated (red)

Fig.8: “Sensorless” test for load torque change

6. CONCLUSIONS

An estimation method of the rotor position, speed and load torque for a surface PMSM has been proposed. It is based on a model that includes the mechanical effect of a high-frequency injection. If the HF injection technique is commonly used in literature, the proposed method is different because it does not rely on saliency of machine. Thus it can be used for either interior or surface PMSM. Furthermore, as the model is observable at standstill, the observer can be applied for position control monitoring or sensorless position control. The high position accuracy is obtained notably for monitoring application. For sensorless position control under

load torque, the accuracy can be improved by using a controller that takes into account the torque changes. Further works could study the development of others non-linear observers, such as sliding mode or high gain observers, instead of EKF.

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